



LLOYDS  
METALS &  
ENERGY  
LIMITED  
OCCUPATIONAL  
HEALTH & REMOTE  
MINING  
CONCESSIONS  
DIVISION – PNG

MASTER  
TECHNICAL  
DOSSIER

INSTITUTIONAL HEALTHCARE INFRASTRUCTURE & FIELD OPERATING SYSTEM

# LLOYDS MEDICAL OS (LMOS-PNG)

A completely autonomous, 100% offline-first clinical hospital and occupational health operating system engineered specifically for remote mineral concessions, bush exploration camps, and rural community health posts across Papua New Guinea. Features live Lead II ECG waveform telemetry, Manchester Emergency Triage, automated date-based patient identification (LMEL-YYYYMMDD-XXX), full printable A4 medical files, automated pharmacy POS dispensation, end-of-day financial balancing with zero-variance locking, triple-sign-off shift audit certification, and military-grade AES-256 encrypted multi-tab corporate archives.

## OPERATING MANDATE & CORE ARCHITECTURE:

Zero Cloud Dependency • Self-Contained SQLite 3 Write-Ahead Logging (WAL) Engine • Multi-Device Wi-Fi LAN Bridge (No WAN required) • 1-Click Plug-and-Play USB Portability (Windows & macOS) • ISO A4 & 80mm Thermal Receipt Print Pipeline • Full Compliance with Papua New Guinea National Department of Health (NDoH) & Mining Safety Standards.

COMMISSIONED FOR  
Executive Leadership &  
Mine Medical Board

TECHNICAL ARCHITECT  
Kritin Rautela (AI/ML Lead)

SYSTEM ARCHITECTURE  
100% Offline SQLite WAL  
+ Multi-Device LAN

DOSSIER RELEASE  
v3.0-PNG Master Edition  
(2026)

01 EXECUTIVE MISSION, OPERATING CONTEXT & PROBLEM STATEMENT

STRATEGIC CONTEXT

Remote mineral exploration concessions, open-pit mines, and extraction assets operated by **Lloyds Metals & Energy Limited (LMEL)** in Papua New Guinea (e.g. Morobe Province, Bulolo Valley, Markham River Basin, and Highland Ridges) face extreme environmental, technical, and human factors challenges:

THE RURAL CONCESSION REALITY

- Zero or Severed Internet:** Highland cloud cover, jungle storms, and generator fuel shortages frequently knock out satellite and cellular links for days or weeks.
- Acute Clinical Hazards:** Remote crews face Falciparum malaria, venomous snakebites (Papuan Taipan / Death Adder), heavy equipment blast/crush trauma, and acute dehydration.

THE OPERATIONAL & FINANCIAL FAILURE

- Standard SaaS Cloud EHRs Fail:** Epic, Cerner, or web portals freeze the moment internet drops, forcing staff back to illegible paper notebooks.
- Inventory & Cash Pilferage:** Without real-time stock-to-patient linkage and daily cash reconciliation, vital antibiotics disappear and clinic revenues vanish.

The **Lloyds Medical Operating System (LMOS-PNG)** was engineered from the ground up to conquer these constraints. It is an entirely self-contained digital hospital platform that runs directly from a high-speed USB flash drive on any clinic PC or laptop with **zero external database installation and zero internet required**.

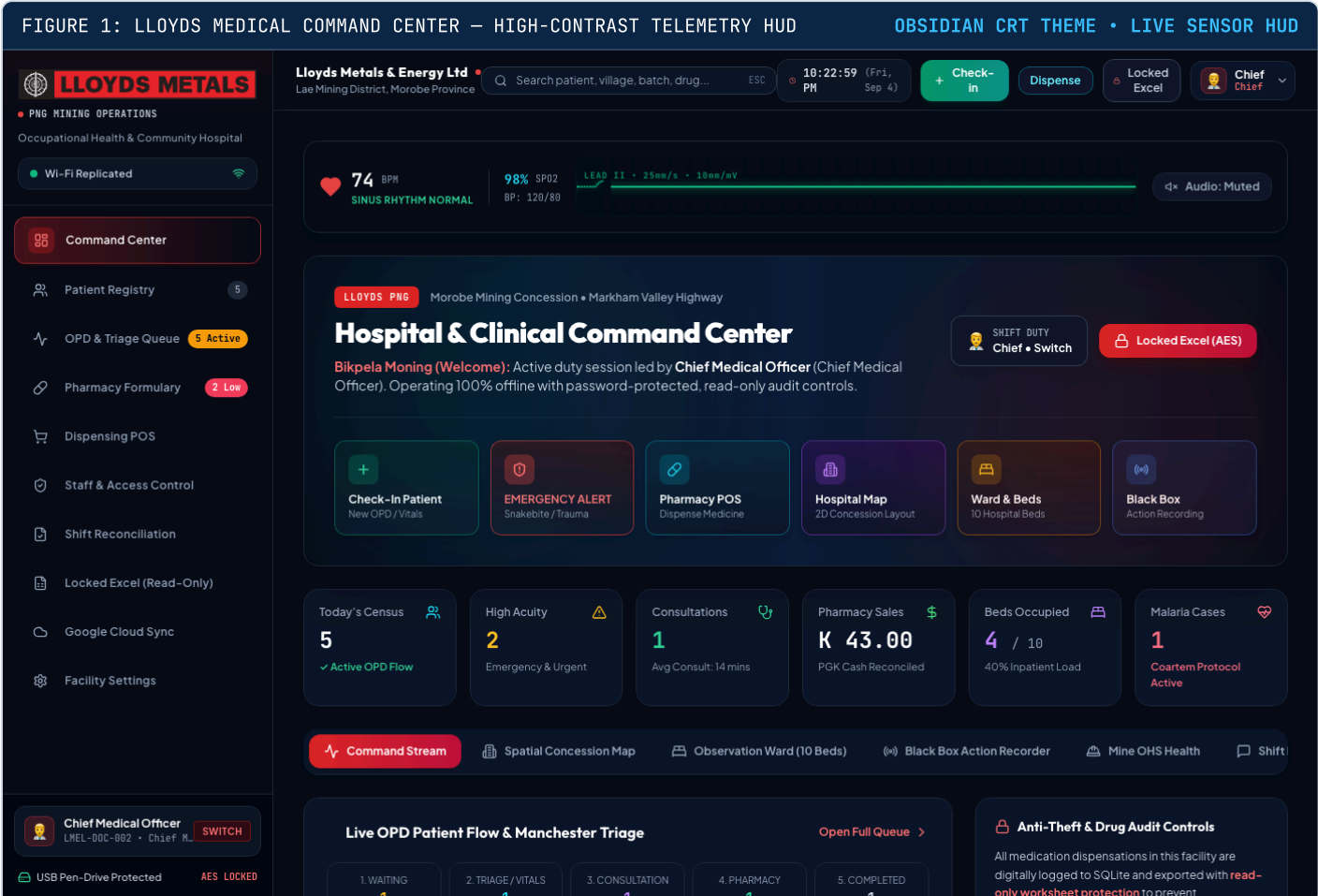


Figure 1: Master Command Center featuring real-time vital signs telemetry (BP 120/80 mmHg, SpO2 98%, HR 74 BPM), Lead II ECG oscilloscope with CRT persistence, 5-stage patient queue counter, weather telemetry, and emergency alert triggers.

02 CORE ENGINEERING ARCHITECTURE & TECHNOLOGY STACK

SYSTEM BLUEPRINT

LMOS-PNG is built upon a hardened, microservices-grade monorepo architecture engineered for extreme durability, instant response times (<10ms per query), and 100% offline autonomy:

| LAYER                   | TECHNOLOGY & COMPONENT                           | ARCHITECTURAL RATIONALE & OPERATIONAL ROLE                                                                                                                                                                |
|-------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend UI             | React 18 • Vite 5 • Tailwind CSS • Lucide Icons  | Zero-lag Single Page Application (SPA). Instant client-side state transitions, responsive high-contrast HUD optimized for low-light clinical environments and field laptops.                              |
| Backend API             | Node.js 18+ • Express.js Microservice            | Local RESTful API handling patient registration, triage queue state machines, inventory math, and financial audits. Listens on <code>http://localhost:4000</code> and LAN interface.                      |
| Database Engine         | SQLite 3.40+ with WAL (Write-Ahead Logging)      | Zero-daemon embedded relational database stored in a single file ( <code>server/data/hospital.db</code> ). WAL mode enables concurrent reads during writes and full resilience against sudden power cuts. |
| Telemetry Engine        | HTML5 Canvas 2D API • Web Audio API              | Hardware-accelerated rendering of real-time Lead II ECG waveforms (25mm/s standard speed) with synthetic sound oscillator simulating medical monitor beeps without external MP3 files.                    |
| Security & Cryptography | Salted SHA-256 Hashes • AES-256 Excel Encryption | Staff passwords hashed with unique salts. Automated generation of password-locked Excel workbooks ( <code>excel4node</code> ) requiring the Chief Medical Officer PIN to access.                          |
| Print Engine            | CSS 3 Paged Media (@page, @media print)          | Browser-native high-fidelity rendering of A4 Clinical Dossiers, Shift Audit Sheets, and 80mm Thermal Receipt rolls with zero external PDF rendering drivers.                                              |

03 COMPLETE RELATIONAL DATABASE ARCHITECTURE (14 MASTER TABLES)

The complete clinical, administrative, and financial state of the hospital is stored in `server/data/hospital.db` across 14 normalized tables. Foreign keys and cascade constraints maintain relational integrity:

| TABLE NAME             | PRIMARY KEY   | KEY FIELDS & DATA TYPES                                                                             | PURPOSE & REGULATORY FUNCTION                                                                                            |
|------------------------|---------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| hospital_settings      | id<br>INTEGER | name, reg_number, doctor_in_charge, currency_symbol, export_password                                | Facility legal accreditation (PNG-MOH-LMEL-2026), Chief Medical Officer credentials, and tax settings.                   |
| patients               | id<br>INTEGER | patient_code (UNIQUE), full_name, age, gender, blood_group, allergies, address_or_village           | Master patient index. Stores date-stamped unique ID (LMEL-YYYYMMDD-XXX), medical history, and landowner clan.            |
| visits                 | id<br>INTEGER | visit_code, patient_id (FK), triage_priority, bp, pulse, spo2, doctor_notes, diagnosis              | Clinical encounters, vital signs telemetry, Manchester triage status, physician diagnoses, and OPD consultation charges. |
| drugs                  | id<br>INTEGER | code (UNIQUE), name, category, unit_price, cost_price, stock_quantity, min_stock_alert, expiry_date | Pharmacy inventory based on Papua New Guinea Essential Medicines. Tracks FEFO expiry and reorder alerts.                 |
| dispensations          | id<br>INTEGER | invoice_number (UNIQUE), patient_id (FK), total_amount, paid_amount, payment_method                 | POS cashier receipts. Links every financial transaction to a patient and clinical encounter.                             |
| dispensation_items     | id<br>INTEGER | dispensation_id (FK), drug_id (FK), quantity, unit_price, subtotal, instructions                    | Itemized medication rows. Automatically decrements inventory and stores exact dosage directions.                         |
| end_of_day_reports     | id<br>INTEGER | report_date (UNIQUE), total_patients, total_opd_fees, total_pharmacy_sales, cash_reconciled         | Daily financial balancing sheet. Records cashier drawer counts, variance, and shift closure lock.                        |
| cloud_sync_config      | id<br>INTEGER | provider, auto_sync_on_wifi, google_sheet_id, webhook_url, last_sync_status                         | Configuration for hybrid cloud sync, Google Sheets integration, and offline buffering triggers.                          |
| sync_logs              | id<br>INTEGER | sync_type, records_count, status, message, synced_at                                                | Complete audit trail of all background sync transactions, packet sizes, and transmission outcomes.                       |
| users                  | id<br>INTEGER | username (UNIQUE), password_hash, role, staff_id, department, status                                | Staff credentials, roles (Doctor, Nurse, Pharmacist, Cashier, Admin), and salted SHA-256 password hashes.                |
| inpatient_beds         | id<br>INTEGER | bed_code (UNIQUE), ward_name, patient_id, acuity_level, diagnosis, attending_doctor                 | Ward bed matrix. Tracks occupancy across Observation Beds, Trauma Bays, and General Wards.                               |
| shift_handover_notes   | id<br>INTEGER | shift_type, author_name, author_role, priority, title, note_content                                 | Clinical handover memos passed between departing and incoming medical staff across 12-hour shifts.                       |
| occupational_incidents | id<br>INTEGER | incident_code (UNIQUE), employee_id, department, incident_type, severity, fit_for_work_status       | Mine occupational injury logs. Tracks MTI, LTI, Near Misses, and corporate fit-for-work certifications.                  |

| TABLE NAME    | PRIMARY KEY   | KEY FIELDS & DATA TYPES                                                      | PURPOSE & REGULATORY FUNCTION                                                                                |
|---------------|---------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| activity_logs | id<br>INTEGER | action_type, user_name, user_role, patient_name, location, details, severity | Tamper-evident flight recorder. Logs every admission, prescription modification, cash entry, and safe eject. |

```
-- Sample SQLite WAL Initialization & Date-Based Patient Code Query
PRAGMA journal_mode = WAL;
PRAGMA synchronous = NORMAL;
PRAGMA foreign_keys = ON;
-- Collision-Proof Sequence Lookup (server/routes/patients.js)
SELECT patient_code FROM patients WHERE patient_code LIKE 'LMEL-20260904-%' ORDER BY patient_code DESC LIMIT 1;
-- Returns: LMEL-20260904-008 → Automatically increments to LMEL-20260904-009
```

Module A: High-Resolution Vital Signs Telemetry & ECG Oscilloscope

The patient telemetry engine calculates vital sign trends in real time. The CRT-style ECG monitor simulates Lead II telemetry running at 25 mm/s with synthetic phosphor trail decay. An integrated audio synthesizer triggers an optional 800Hz / 400Hz acoustic beep on each QRS peak to replicate intensive care cardiac monitors.

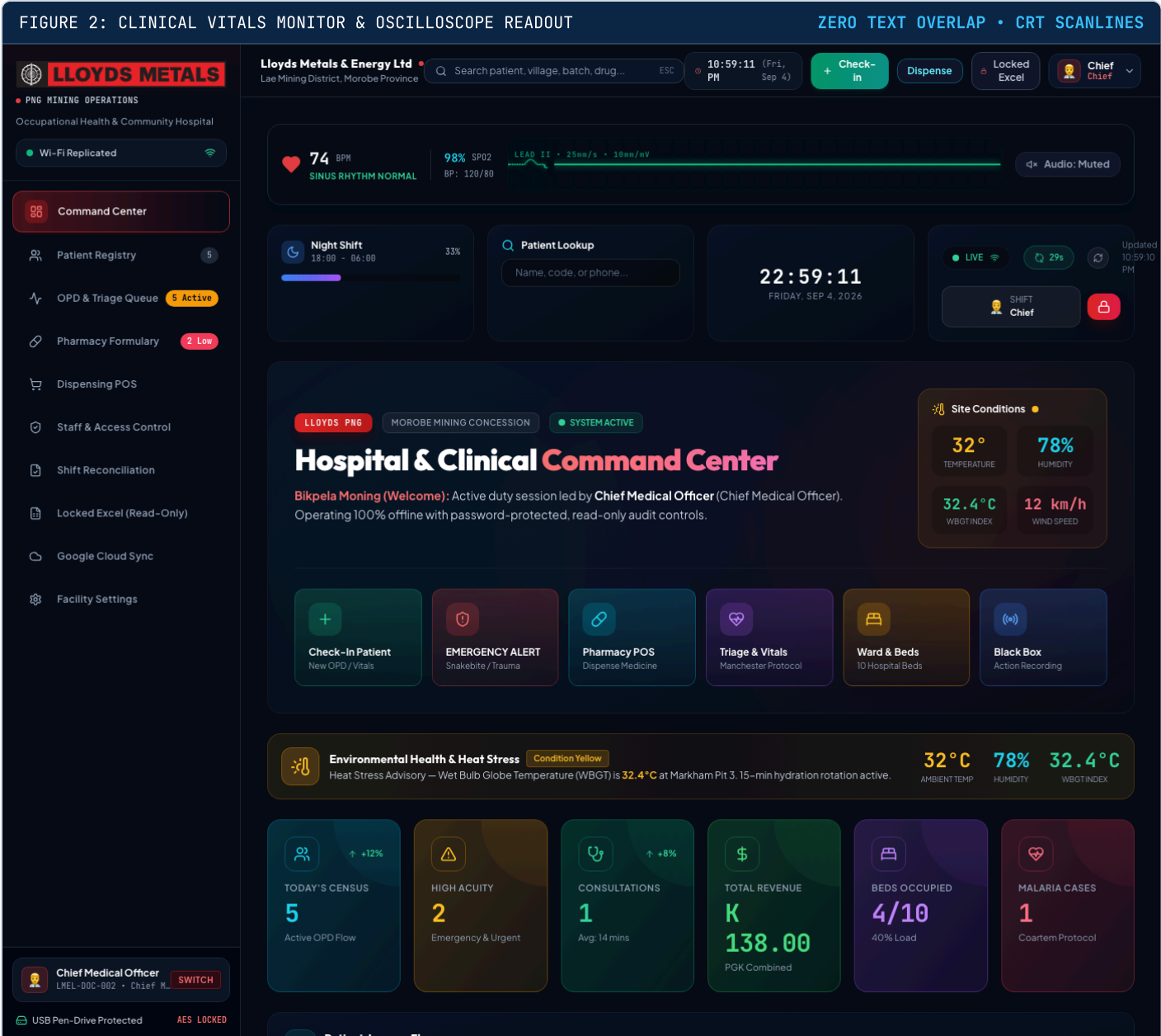


Figure 2: Verified vital signs telemetry tile showing crystal-clear typography for Blood Pressure (120/80 mmHg), SpO2 (98%), Heart Rate (74 BPM), and Respiratory Rate (16/min).

Module B: Date-Stamped Patient Registry (LMEL-YYYYMMDD-XXX)

To eliminate patient file confusion in remote areas where workers and village members may share similar names, the system automatically assigns an immutable date-stamped identifier upon check-in: LMEL-[YEAR] [MONTH] [DAY] - [SEQUENCE] (e.g. LMEL-20260904-008). This guarantees chronological traceability across years of operations.

Figure 3: Live Patient Registry showing date-based identifiers (LMEL-20260904-008, LMEL-20260904-007), patient demographics, village settlements, active allergy tags (Penicillin, Aspirin), and instant 1-click Printable Medical Dossier button.

Queries the maximum existing sequence for today's date and increments atomically. If registration occurs offline across shifts, no duplicate IDs can ever be generated.

The blue Eye button opens the complete A4 printable patient dossier, aggregating longitudinal vital signs, doctor notes, and full pharmacy dispensation history.



05 CLINICAL WORKFLOW: OPD TRIAGE, QUEUE & DOCTOR CONSULTATIONS

PATIENT JOURNEY

The Outpatient Department (OPD) queue coordinates the entire patient journey through a 5-stage clinical state machine: Waiting → Triage & Vitals → In Consultation → At Pharmacy → Completed / Admitted.

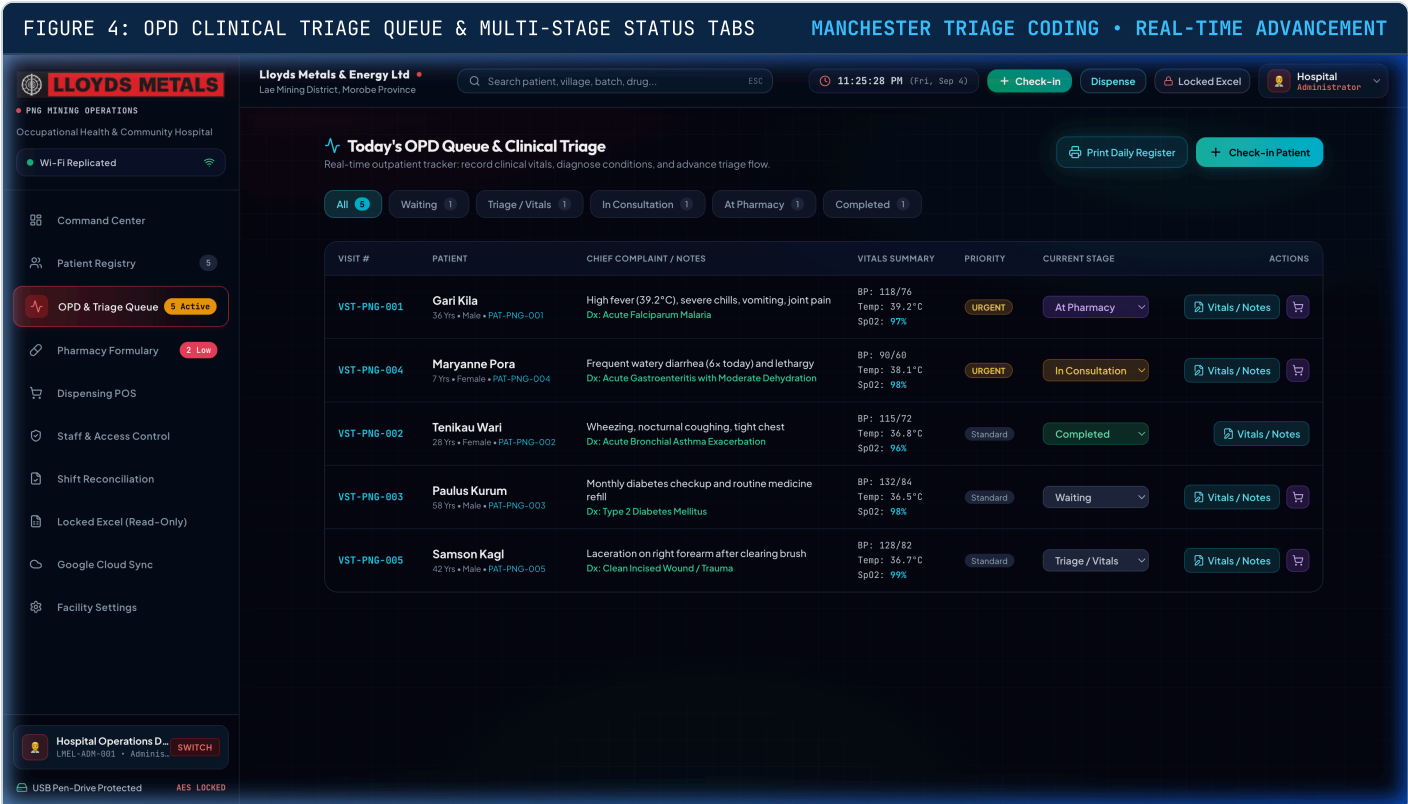


Figure 4: OPD clinical queue displaying 5 active patient stages, Manchester Triage urgency tags (Orange = Urgent, Yellow = Standard), chief complaints, vital signs, and immediate consultation action buttons.

06 MODULE C: PHARMACY FORMULARY & POINT-OF-SALE (POS) DISPENSATION

INVENTORY & PHARMACY

The pharmacy formulary comes pre-loaded with over 50 essential medications tailored to Papua New Guinea health standards. Every dispensation requires a linked clinical consultation or patient file, preventing unrecorded stock leakage:



**FEFO EXPIRY CONTROL • LOW STOCK ALARMS**



Figure 6: POS dispensing interface allowing rapid medication selection, dosage instruction entry, cash collection, and instant generation of bilingual receipts.

LMOS-PNG features an integrated vector cartography GIS engine plotting patient origins, mine worksites, and landowner villages. It enables medical officers to detect disease outbreaks (e.g. malaria clusters) and plan emergency medevac extraction routes:

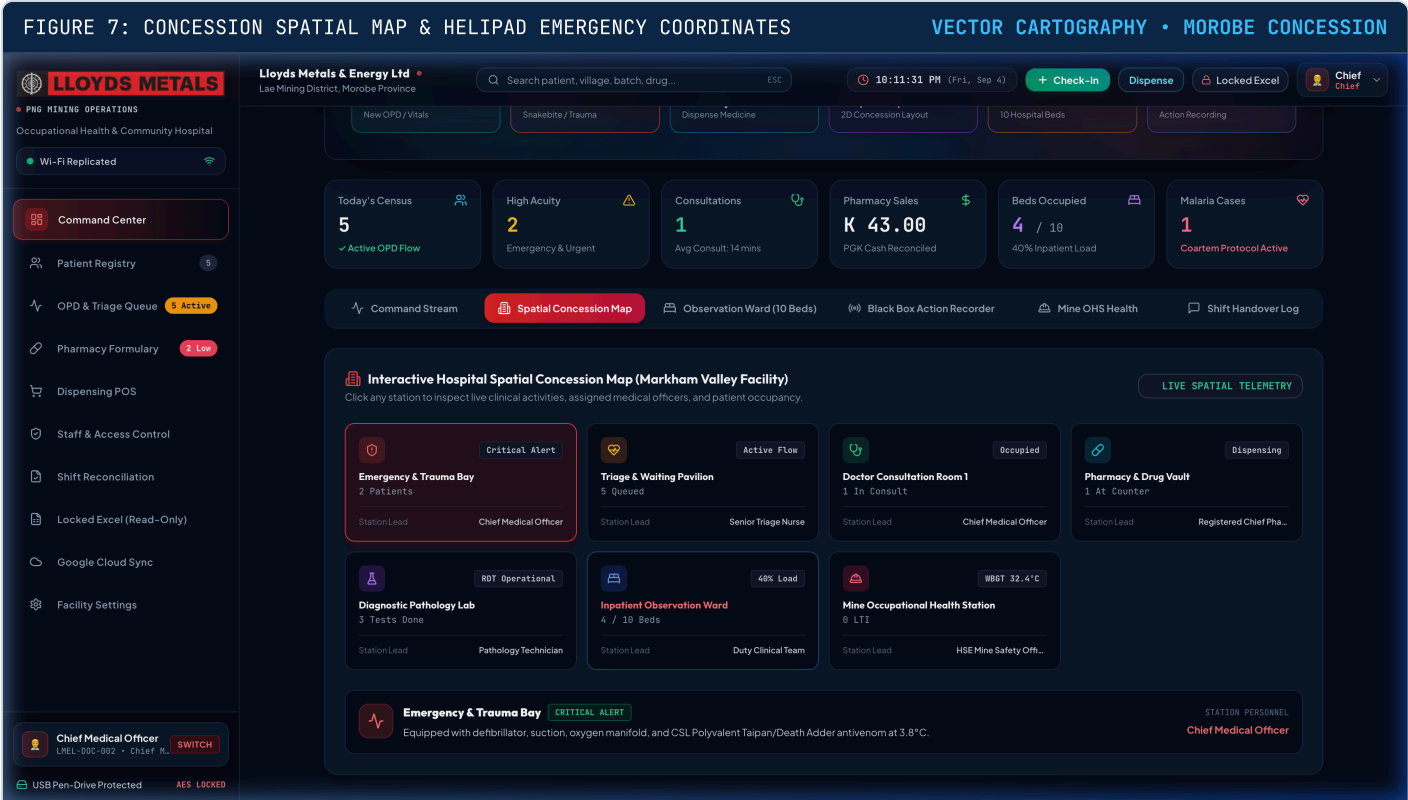
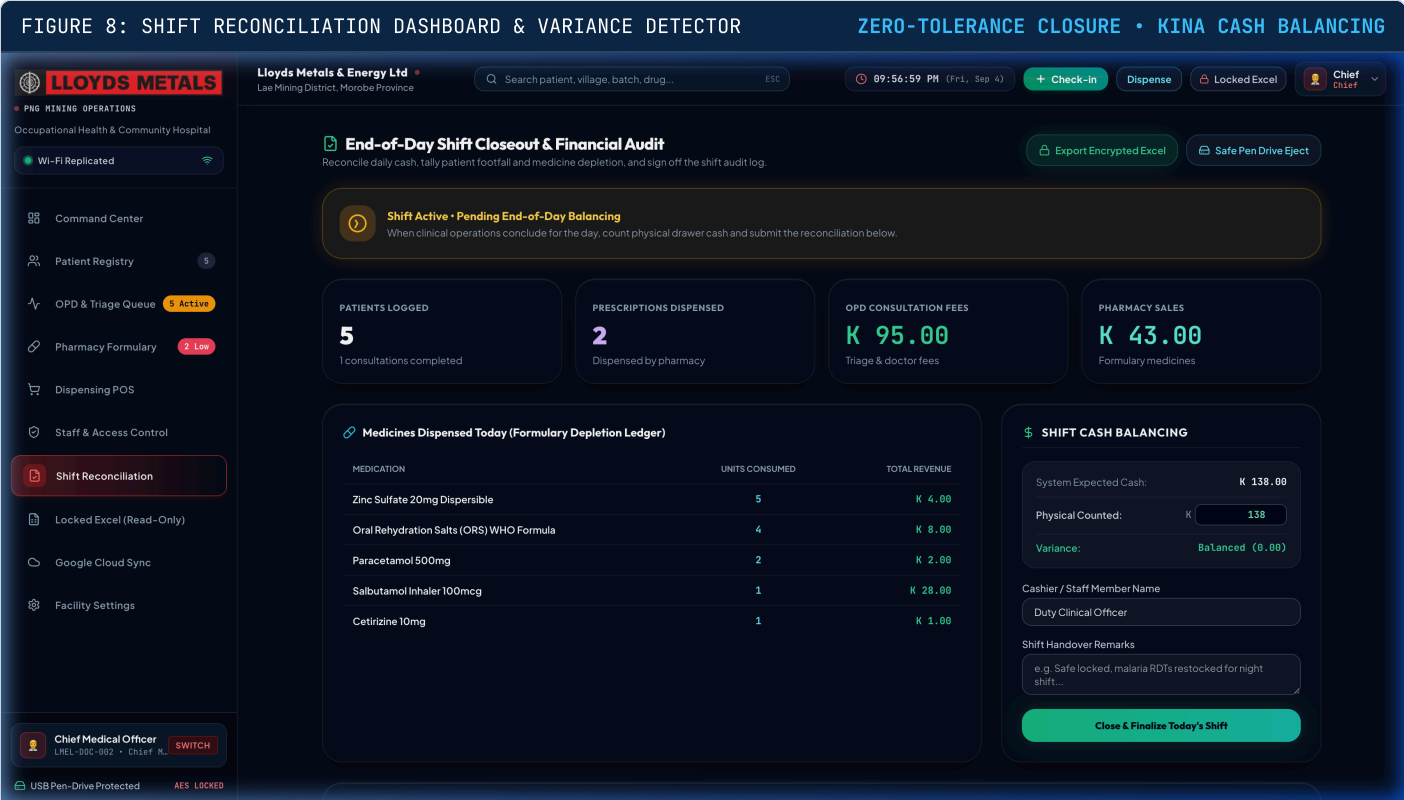


Figure 7: Concession spatial map plotting active cases across Mine Pit Alpha, Exploration Camp 1, Markham Valley, and emergency evacuation Helipad Hub (06°44'S, 146°59'E).

At the conclusion of each duty shift, the system enforces a strict cash drawer balancing routine. The software automatically calculates expected revenue from OPD fees and pharmacy sales, and compares it with physical cash counted in the safe:



09 MODULE F: INSTITUTIONAL SHIFT AUDIT & TRIPLE SIGN-OFF  
CERTIFICATE

MEDICO-LEGAL  
AUDIT

Closing a shift generates an official A4 Audit Certificate designed to satisfy internal mine auditors, insurance underwriters, and Papua New Guinea National Department of Health inspectors:

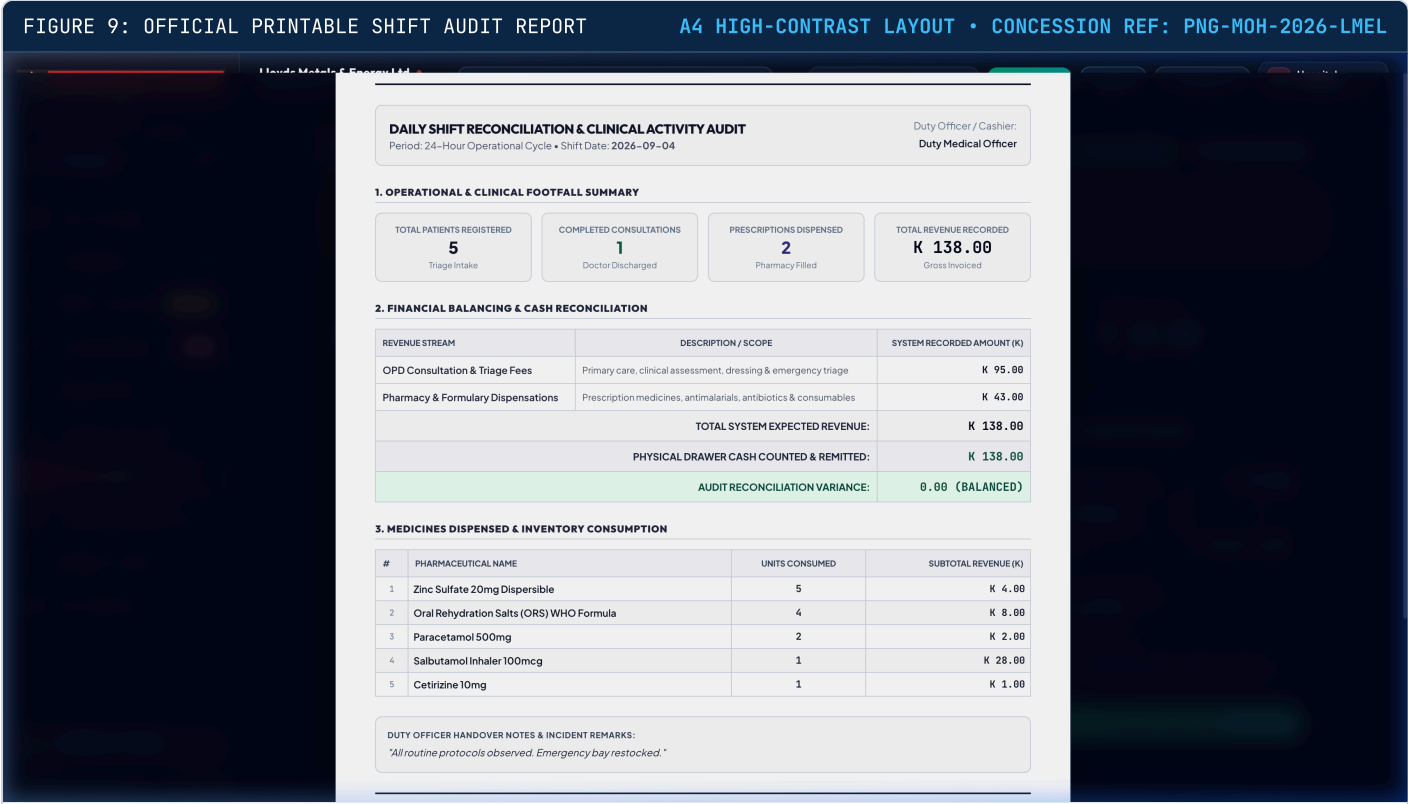


Figure 9: Official Daily Shift Reconciliation & Clinical Activity Audit Sheet displaying executive footfall summary, financial breakdown (OPD fees + pharmacy sales), and inventory depletion table.

FIGURE 10: TRIPLE SIGNATURE & CONCESSION SEAL  
CERTIFICATION

CHIEF MEDICAL OFFICER • TRIAGE NURSE • CONCESSION  
STAMP

Lloyds Metals & Energy Ltd

| REVENUE STREAM                           | DESCRIPTION / SCOPE                                              | SYSTEM RECORDED AMOUNT (K) |
|------------------------------------------|------------------------------------------------------------------|----------------------------|
| OPD Consultation & Triage Fees           | Primary care, clinical assessment, dressing & emergency triage   | K 95.00                    |
| Pharmacy & Formulary Dispensations       | Prescription medicines, antimalarials, antibiotics & consumables | K 43.00                    |
| TOTAL SYSTEM EXPECTED REVENUE:           |                                                                  | K 138.00                   |
| PHYSICAL DRAWER CASH COUNTED & REMITTED: |                                                                  | K 138.00                   |
| AUDIT RECONCILIATION VARIANCE:           |                                                                  | 0.00 (BALANCED)            |

3. MEDICINES DISPENSED & INVENTORY CONSUMPTION

| # | PHARMACEUTICAL NAME                      | UNITS CONSUMED | SUBTOTAL REVENUE (K) |
|---|------------------------------------------|----------------|----------------------|
| 1 | Zinc Sulfate 20mg Dispersible            | 5              | K 4.00               |
| 2 | Oral Rehydration Salts (ORS) WHO Formula | 4              | K 8.00               |
| 3 | Paracetamol 500mg                        | 2              | K 2.00               |
| 4 | Salbutamol Inhaler 100mcg                | 1              | K 28.00              |
| 5 | Cetirizine 10mg                          | 1              | K 1.00               |

DUTY OFFICER HANDOVER NOTES & INCIDENT REMARKS:  
"All routine protocols observed. Emergency bay restocked."

OFFICIAL CLINICAL & ADMINISTRATIVE AUTHORIZATION CERTIFICATION

DUTY DISPENSER / CASHIER

Duty Medical Officer

Signature & Date

SUPERVISING MEDICAL OFFICER

Chief Medical Officer (Occupational & Tropical Medicine)

Signature & Date

OFFICIAL CLINICAL STAMP

Affix Lloyds Metals Concession Stamp Here

VERIFIED & ENCRYPTED

Generated by Lloyds Metals Occupational Health System v2.6 (Offline-First Edition)

Printed on: 9/4/2026, 11:23:29 PM

Concession Ref: PNG-MOH-2026-LMEI

Figure 10: Certification footer on A4 print audit report with designated sign-off lines for Duty Dispenser, Supervising Medical Officer, and official Lloyds Metals Concession Circular Stamp Box.

THE TRIPLE SIGN-OFF LEGAL GUARANTEE

A shift report is not legally finalized until physically reviewed and signed by:

- 1. **Duty Dispenser / Pharmacist:** Certifies all stock decrements match physical medicine packages given to patients.
- 2. **Supervising Medical Officer (Doctor):** Certifies clinical diagnoses, patient admission decisions, and triage accuracy.
- 3. **Official Concession Stamp:** Affixed in the designated stamp box to prevent counterfeit report substitution.

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MODULE G: MEDICO-LEGAL BLACK BOX ACTIVITY RECORDER

TAMPER-PROOF AUDIT TRAIL

To prevent administrative fraud and provide tamper-evident proof during workplace injury inquiries, LMOS-PNG records every critical event into an append-only SQLite log table (activity\_logs):

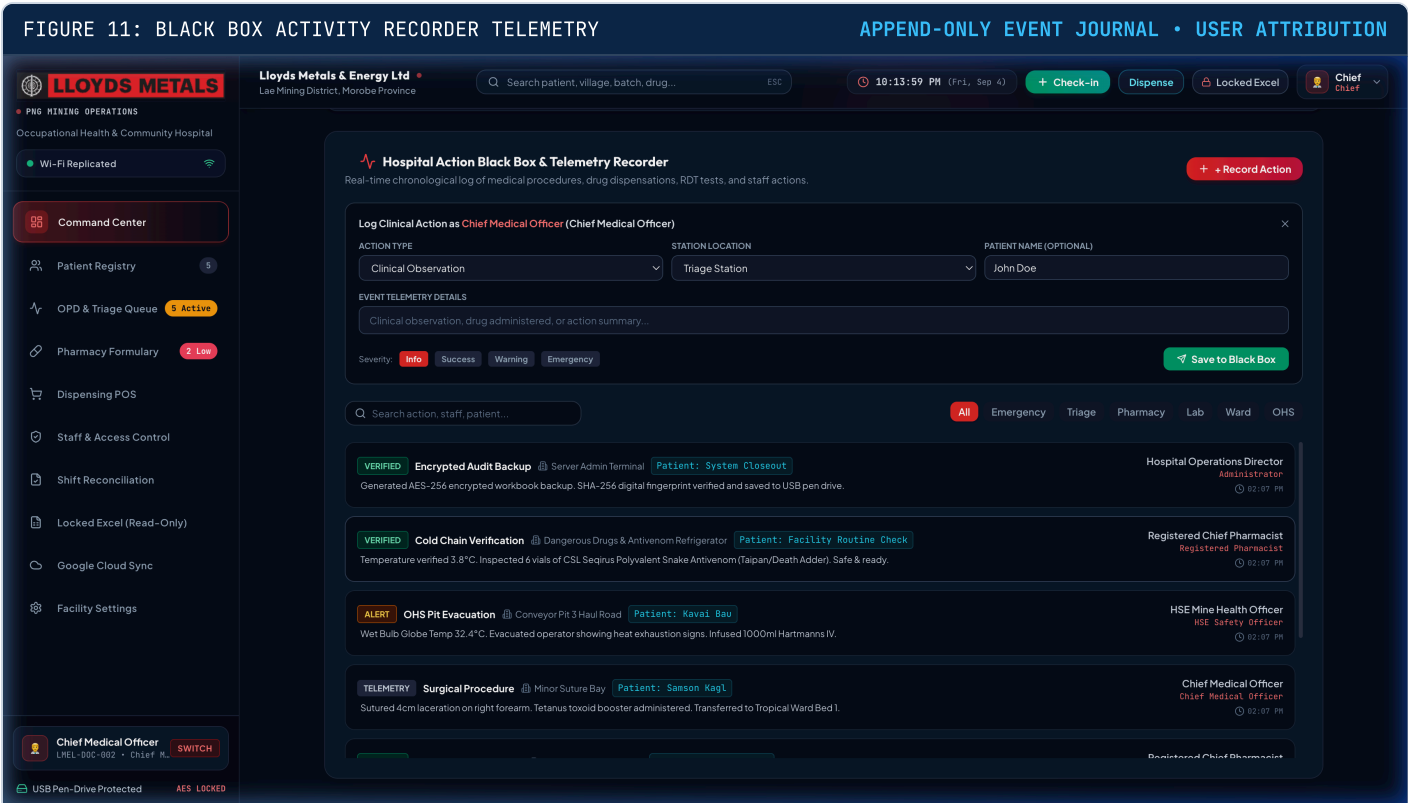


Figure 11: Medico-legal activity telemetry logging patient intake timestamps, clinical consultations, drug dispensations, and system exports with user attribution and severity tagging.

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MODULE H: MULTI-ROLE STAFF ACCESS CONTROL & CRYPTOGRAPHY

ACCESS SECURITY

The platform enforces strict role-based access control (RBAC). Passwords are never stored in plain text; they are encrypted using salted SHA-256 hashes:



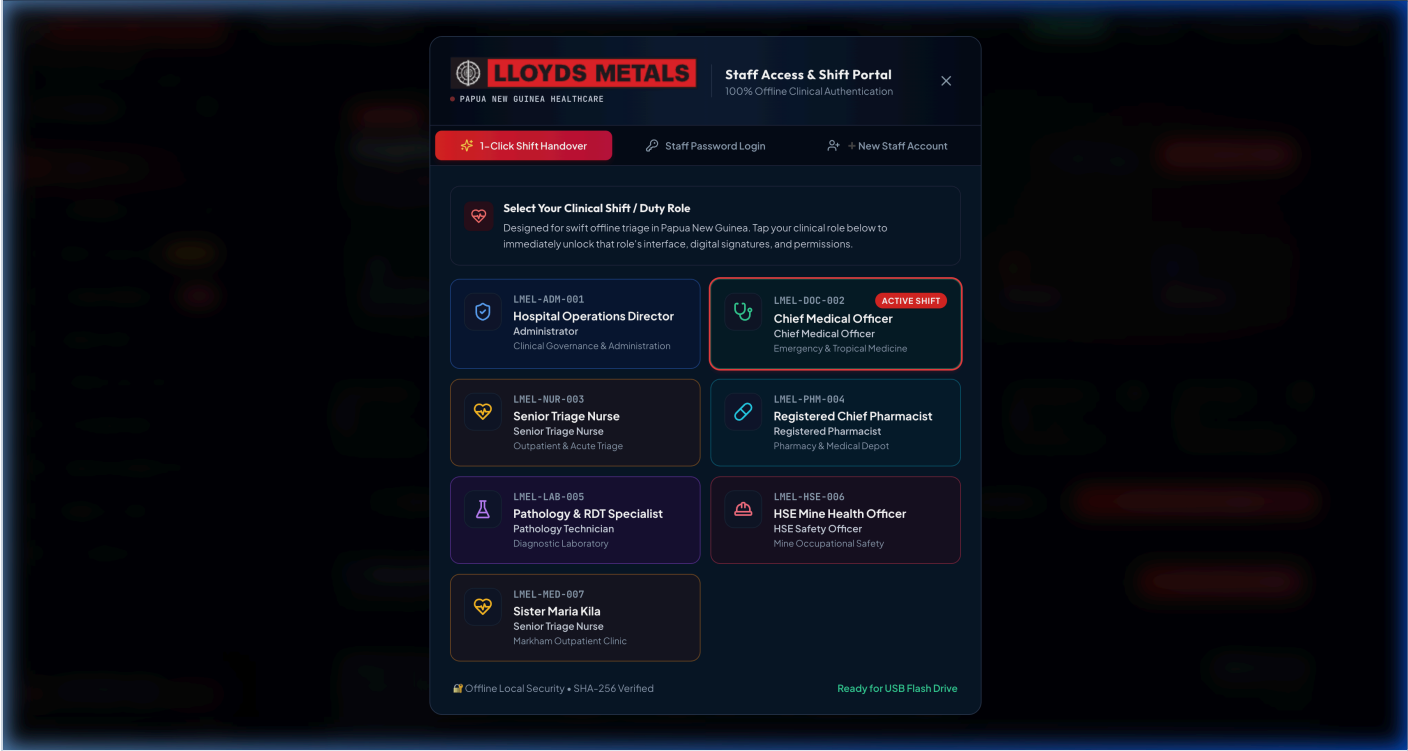


Figure 12: Offline staff authentication modal supporting rapid role switching (Doctor, Nurse, Pharmacist, Cashier, Admin) and credential updates without internet connection.

| ROLE                   | DEFAULT USERNAME | DEFAULT PASSWORD | AUTHORIZATIONS & PRIVILEGES                                                                         |
|------------------------|------------------|------------------|-----------------------------------------------------------------------------------------------------|
| Master Admin Console   | admin            | lloyds2026       | Full system control, staff management, password changes, and hospital settings.                     |
| Chief Medical Officer  | doctor           | lloyds2026       | Clinical diagnosis, prescription authorization, inpatient bed admissions, and shift audit sign-off. |
| Senior Triage Nurse    | triage_officer   | lloyds2026       | Patient registration, Manchester triage scoring, vital signs entry, and trauma timers.              |
| Pharmacy Officer       | pharmacist       | lloyds2026       | Prescription dispensing, batch inventory tracking, and POS cashier receipts.                        |
| Hidden Excel Admin PIN | (Unlock PIN)     | lloyds2026       | Unlocks encrypted multi-sheet Excel generator for corporate board exports.                          |

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MODULE I: MILITARY – GRADE AES – 256 ENCRYPTED MULTI – TAB EXCEL EXPORT

CORPORATE  
REPORTING

When clinic data must be transmitted to corporate headquarters or government auditors, LMOs-PNG generates a password-protected multi-worksheet Excel document (exce14node) requiring the PIN 11loyds2026 to decrypt:

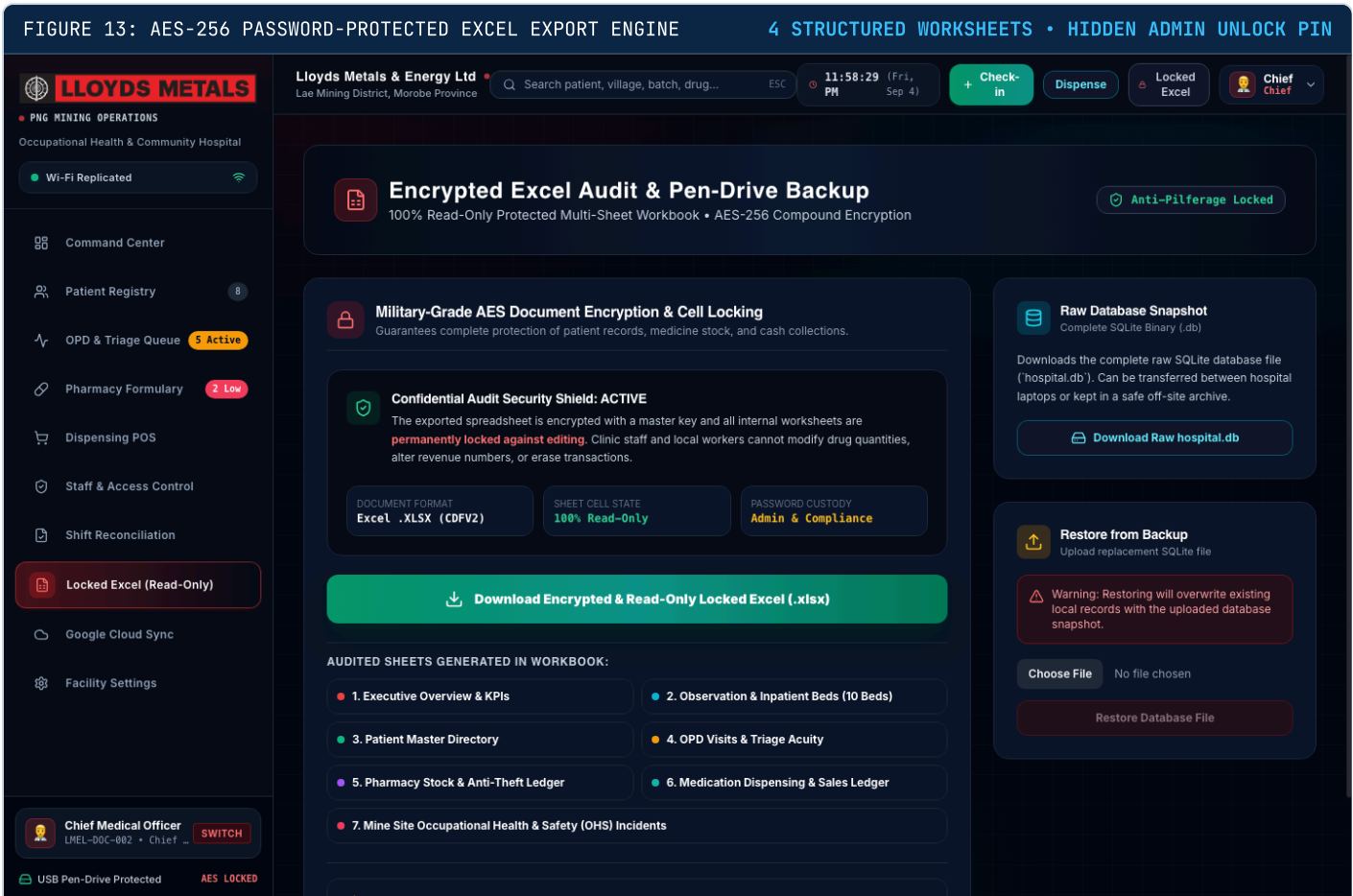


Figure 13: Encrypted Excel export interface showing PIN-protected unlock mechanism and automatic generation of multi-sheet workbooks with formulas and conditional formatting.

Sheet 1: Executive Summary

Total patients treated, daily consultation revenue, pharmacy sales, inpatient bed occupancy percentage, and critical low-stock alerts.

Sheet 2: Clinical Encounters

Complete patient visit logs including date-based IDs, blood pressures, oxygen saturations, physician notes, and final diagnoses.

Sheet 3: Pharmacy Inventory

Complete pharmaceutical stock status: physical quantity on hand, unit cost, selling price, FEFO expiry dates, and reorder alerts.

Sheet 4: Cash Drawer Journal

Itemized financial audit of all cash receipts, consultation fees, cashier attributions, and drawer variance figures.

13

MODULE J: HYBRID GOOGLE CLOUD & LAN RELAY SYNCHRONIZATION

HYBRID SYNC  
RELAY

The platform operates with an intelligent network sensor. While completely offline, it writes transactions to a local SQLite queue. The moment a satellite terminal or cellular 4G hotspot is detected, it automatically transmits summarized telemetry:

Figure 14: Cloud sync dashboard displaying live connection status, Google Sheets spreadsheet linkage, auto-sync triggers, and timestamped sync audit history.

## 14 FIELD DEPLOYMENT TOPOLOGIES & HARDWARE SETUP

FIELD SPECIFICATIONS

LMOS-PNG supports three distinct field deployment topologies depending on the physical infrastructure available at the concession station:

| DEPLOYMENT MODEL                     | HARDWARE & WIRING SETUP                                                                                                     | NETWORK TOPOLOGY                                                     | OPERATIONAL BEST FIT                                                                                          |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>1. Portable Pen Drive (USB)</b>   | Any standard Windows or Mac PC. System runs directly off a 16GB/32GB Kingston/SanDisk flash drive.                          | 100% Offline (No router, no Wi-Fi, zero internet required).          | Mobile drill-rig exploration camps, wilderness patrol teams, and emergency medevac stations.                  |
| <b>2. Clinic Local Wi-Fi (LAN)</b>   | 1 primary clinic PC running the server + 1 standard Wi-Fi router (e.g. TP-Link / Netgear). Other tablets connect to router. | Local clinic Wi-Fi (LAN only; router does NOT need WAN or internet). | Multi-room base hospitals where Triage Nurse, Doctor, and Pharmacist use separate tablets/PCs simultaneously. |
| <b>3. Concession Head Office VPS</b> | Dedicated Linux VM (Ubuntu 22.04 LTS / Docker container) running on camp server or cloud.                                   | Camp intranet or VSAT broadband connection.                          | Centralized head-office monitoring across multiple mining concessions in Papua New Guinea.                    |

## 15 STANDARD OPERATING PROCEDURES (CLINIC SOP RUNBOOK)

DAILY CLINIC SOP

### PHASE 1: 06:00 MORNING SHIFT OPENING

1. Insert USB Pen Drive and double-click `Start_Hospital_Windows.bat`.
2. Open Chrome to `http://localhost:4000`.
3. Log into Command Center with Doctor/Nurse credentials.
4. Review Pharmacy Formulary for low-stock alarms.

### PHASE 2: PATIENT INTAKE & TRIAGE

1. Click "Quick Check-in" for new or returning patient.
2. System generates unique ID LMEL-YYYYMMDD-XXX.
3. Enter vital signs (BP, SpO2, HR, Temp) & assign Manchester triage priority.
4. Patient automatically appears in Doctor OPD Queue.

### PHASE 3: DOCTOR CONSULTATION

1. Select patient from active OPD Queue.
2. Review vitals trend and enter clinical notes & diagnosis.
3. Prescribe essential medications with dosage instructions.
4. Click "Save & Route to Pharmacy".

### PHASE 4: PHARMACY DISPENSATION & POS

1. Open Dispense POS and select patient name.
2. Confirm prescribed drugs, quantities, and batch numbers.
3. Collect fee in PNG Kina (PGK) and print bilingual invoice receipt.
4. Stock quantities decrement automatically in real time.

### PHASE 5: 18:00 EVENING SHIFT CLOSEOUT & SAFE EJECT

1. Navigate to **End-of-Day** tab.
2. Count physical cash in safe/drawer and enter total.
3. Confirm variance reads exactly **0.00 BALANCED**.
4. Enter handover notes and click **"Close & Finalize Today's Shift"**.
5. Click **"Print Shift Audit Report"**, sign, and affix official concession circular stamp.
6. Click **"Safe Pen Drive Eject"** to flush SQLite WAL buffers and safely unplug the drive.

The complete source tree is cleanly partitioned between the client Single Page Application, the Express microservice backend, and the embedded SQLite database:

```
/Users/kritin/LLOYDS SERVER / |— Start_Hospital_Windows.bat # 1-Click launcher for Windows PCs |—
Start_Hospital_Mac.command # 1-Click launcher for macOS laptops |— Safe_Pen_Drive_Eject.bat # Safe SQLite flush & eject
for Windows |— Safe_Pen_Drive_Eject.command # Safe SQLite flush & eject for macOS |— README.md # Master engineering &
operational specification |— docs/ | |— Executive_Report_Lloyds_Medical_OS.html # Master Executive Report (Source) |
|— Lloyds_Medical_OS_Executive_Manual.pdf # Compiled Master Executive PDF Manual | |— assets/ # 18 high-resolution
system screenshots |— client/ # Frontend React 18 + Vite SPA | |— index.html # HTML entry point with Lloyds title &
icons | |— vite.config.js # Build configuration (proxy to :4000) | |— tailwind.config.js # Institutional HUD color
palette tokens | |— dist/ # Compiled production bundle served by Express | |— src/ | |— App.jsx # Main router & tab
state manager | |— context/AuthContext.jsx # Multi-role authentication & session state | |— services/api.js # REST
client communicating with local backend | |— components/ | | |— Navbar.jsx # Global header, quick search & modal
buttons | | |— Sidebar.jsx # Tab navigation & live Wi-Fi connection indicator | | |— AuthModal.jsx # Login & staff
account creation modal | | |— QuickCheckInModal.jsx # Patient check-in & triage launcher | | |—
PrintablePatientRecordModal.jsx # Full A4 medical dossier modal | | |— PrintableShiftReportModal.jsx # Triple sign-off
audit modal | | |— PrintableReceiptModal.jsx # Thermal & A4 invoice receipt | |— pages/ | |— Dashboard.jsx # Command
Center HUD & live ECG oscilloscope | |— Patients.jsx # Date-stamped patient registry & search | |— OPDQueue.jsx # 5-
stage clinical triage & consultation queue | |— Pharmacy.jsx # Formulary stock, batch & expiry control | |—
DispensePOS.jsx # POS cashiering & prescription dispensation | |— StaffManagement.jsx # User accounts, roles & SHA-256
passwords | |— EndOfDay.jsx # Shift financial reconciliation & cash balancing | |— ExcelExport.jsx # AES-256 encrypted
multi-tab Excel export | |— CloudSync.jsx # Google Sheets sync & offline buffer status | |— Settings.jsx # Facility
accreditation & tax configuration |— server/ # Backend Node.js Express REST API |— server.js # Express app entry &
static bundle host |— db.js # SQLite3 WAL initialization & query helpers |— data/ | |— hospital.db # Single-file
SQLite database (14 master tables) | |— routes/ | |— auth.js # Login & password hashing endpoints | |— patients.js # Patient
CRUD & LMEL-YYYYMMDD-XXX ID generator | |— visits.js # Clinical encounters & triage queues | |— drugs.js # Pharmacy
formulary & stock depletion | |— dispense.js # Cashier invoices & itemized sales | |— endOfDay.js # Daily balancing &
variance calculation | |— export.js # Encrypted Excel workbook builder (excel4node) | |— dashboard.js # Real-time telemetry
& clinical statistics | |— cloudSync.js # Google Sheets relay & sync logs | |— settings.js # Legal registration & hospital
profile
```

The **Lloyds Medical Operating System (LMOS-PNG)** is the intellectual and operational property of **Lloyds Metals & Energy Limited (LMEL)**. It is deployed strictly for occupational healthcare delivery, emergency triage, and customary landowner health assistance within authorized exploration and mining concessions in Papua New Guinea.

OFFICIAL CONCESSION AUDIT & TECHNICAL SIGN-OFF

PREPARED & CERTIFIED BY:

Kritin Rautela

Technical Lead & AI/ML Specialist

Signature: \_\_\_\_\_

APPROVED BY MEDICAL BOARD:

Chief Medical Officer

MBBS, Head of Concession Health

Signature: \_\_\_\_\_

OFFICIAL CONCESSION SEAL:

